

Project Report: Home Lab Security Hardening - Cybersecurity Analyst

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Family: Cybersecurity Analyst
Level: Beginner
Chapters: 6
Source: CodeQuest project specification (official docs + industry patterns)

Summary

Harden a Linux VM: firewall, SSH keys, fail2ban, updates, and document baseline checklist.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Beginner]

Chapter focus: Project Overview** *(Level: Beginner)

Home Lab Security Hardening** is a beginner-level end-to-end project for Cybersecurity Analyst. **Duration:** 2 weeks. **Harden a Linux VM:** firewall, SSH keys, fail2ban, updates, and document baseline checklist. **Tools:** Linux, UFW, fail2ban, nmap.

Code Reference:

```
Objective: UFW firewall rules configured
Objective: SSH key-only auth (no passwords)
Objective: fail2ban installed
Objective: Security baseline checklist completed
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Cybersecurity Analyst portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Cybersecurity Analyst resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Baseline [Beginner]

Chapter focus: Phase: Baseline** *(Level: Beginner)

Document current state. Complete these tasks: Open ports scan; User accounts. Deliverable for this phase: Baseline audit.

Code Reference:

- [] Open ports scan
- [] User accounts

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Baseline is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Cybersecurity Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Baseline audit to the teacher.

Chapter 3: Phase: Harden [Beginner]

Chapter focus: Phase: Harden** *(Level: Beginner)

Apply CIS basics. Complete these tasks: UFW; SSH keys; fail2ban. Deliverable for this phase: Hardened VM.

Code Reference:

- [] UFW
- [] SSH keys
- [] fail2ban

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Harden is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Cybersecurity Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Hardened VM to the teacher.

Chapter 4: Phase: Verify [Beginner]

Chapter focus: Phase: Verify *(Level: Beginner)**

Re-scan after hardening. Complete these tasks: nmap scan; SSH test. Deliverable for this phase: Before/after comparison.

Code Reference:

- [] nmap scan
- [] SSH test

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Verify is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Cybersecurity Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Before/after comparison to the teacher.

Chapter 5: Phase: Document [Beginner]

Chapter focus: Phase: Document *(Level: Beginner)**

Runbook. Complete these tasks: Steps; Rollback. Deliverable for this phase: Hardening runbook PDF.

Code Reference:

- [] Steps
- [] Rollback

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Document is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Cybersecurity Analyst agile sprints and coursework milestones.

Real use example

Team demo shows the Hardening runbook PDF to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Beginner]

Chapter focus: Final Deliverables and Review** *(Level: Beginner)

Submit all final deliverables: Baseline audit; Hardening runbook; Scan comparison. Skills demonstrated: Linux security, Hardening. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Baseline audit
Deliverable: Hardening runbook
Deliverable: Scan comparison
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Baseline audit and passes the rubric.